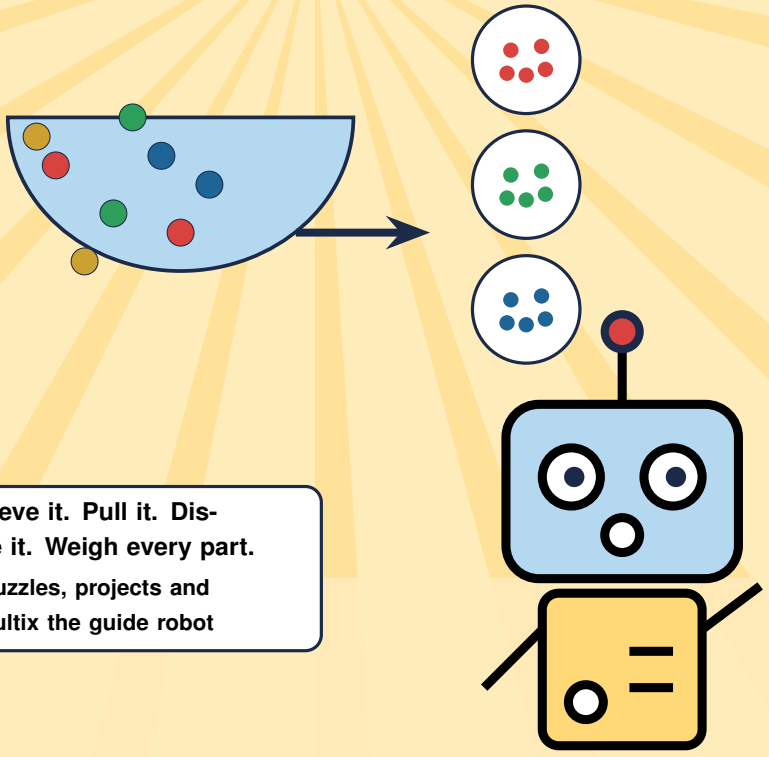
SCIENCE
Book 6

Mixed Up and Sorted Out



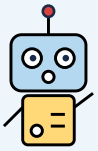
Sieve it. Pull it. Dissolve it. Weigh every part.

Puzzles, projects and Multix the guide robot

Ages 9 to 11

This book belongs to _____

Hello. I am Multix.

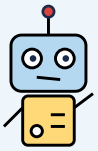


I am a guide robot. I read every instruction very carefully, which is why I sometimes get things wrong. When I make a mistake, see whether you can catch it. I never give away a puzzle answer.

How this book works

Every book in this series asks the same few questions, so once you have read this page you will know your way around.

- **Try it** boxes tell you what to do with a ruler, a cup, some string or a clock. You need nothing from a shop. Everything is at home or at school.
- **What did we keep?** After you simplify something, such as keeping the same conditions in a test or averaging several throws, you say what stayed and what was set aside. Simplifying is fine. Forgetting what you set aside is not.
- **What makes this correction trustworthy?** When you fix or improve an answer, a box says what you checked, what the check gives you reason to trust, and what it does *not* show.
- **Circles** show how a number was made. Page 4 explains how to read them, and Book 1 introduced them.
- **Puzzles** come at the end of each chapter. Try them before you look at the answers on page 45.
- **Slop check** is a page near the end. Slop is hasty, gaudy work that forgot to take care. You will learn to spot it and to fix it.



Here is a fact about me. I stirred a spoonful of salt into water and it vanished. I weighed the cup and decided it was lighter than before. Chapter 3 will explain why I should be embarrassed.

In this book

1	What a Mixture Is	5
2	Sieves and Magnets	10
3	Salt That Disappears	15
4	Muddy Water Through Paper	21
5	Separating Everything	27
6	Finding What Went Missing	32
⋆	Slop check	38
⋆	Answers	45
⋆	Words, and notes for grown-ups	47

Try it: Gather your kit

You will use these things across the six chapters. Everything is found at home or at school. Measuring comes from Book 1, fair testing from Book 2 and the ledger from Book 5, and all three come back here.

- a kitchen scale in grams
- three clear cups
- a spoon and a saucer
- a sieve or tea strainer
- a magnet and paper clips
- coffee filters or paper towel
- table salt and some sand
- rice and lentils
- a little garden soil
- pencil and notebook

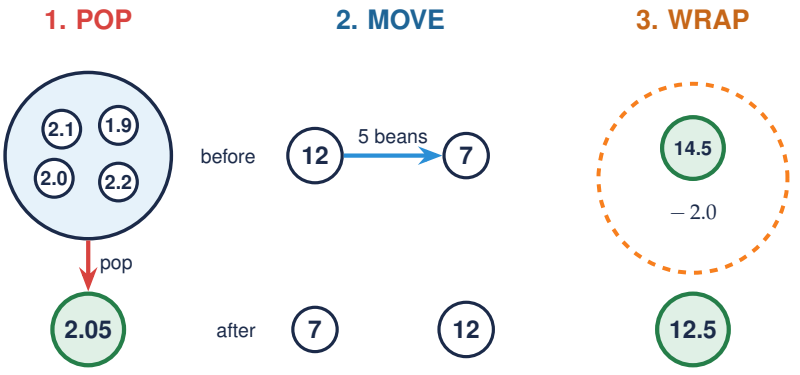
■ **Adult help:** Never taste anything from an experiment, not even the salt water. Keep small magnets away from little children and pets, since swallowed magnets are dangerous. Wipe up spills. Nothing here needs a flame or mains electricity.

Units in this book

We use the metric system: centimetres (cm), metres (m) and seconds (s). For mixtures we use grams (g) on a kitchen scale. A scale that reads to 1 g can be off by about 1 g, and we will keep that in mind. A paper clip weighs about 1 g. There are 10 mm in 1 cm and 100 cm in 1 m. If you are used to inches, one inch is about 2.5 cm. We will use that only in side notes.

Reading the circles

A circle holds a group of things together. There are three things you can do with circles in this series.

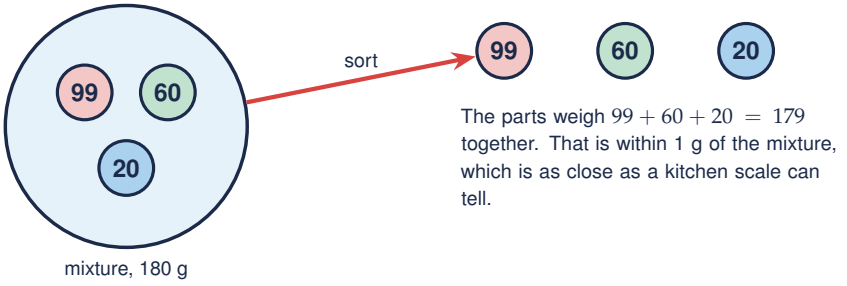


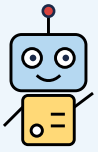
Pop Squash a group down to one value that sums it up, such as an average. A pop is a *summary*, so it always loses something. Every time we pop, we say what the summary keeps and what it loses.

Move An arrow that carries something from one circle to another, such as beans or water. Nothing is summarized. You will see more of moves in later books.

Wrap When you correct something, draw a new dashed circle around the old one and show what changed. The old work stays visible, so anyone can check it.

A mixture is a circle holding smaller circles





A pop is not a lie, but it is a shortcut. Any time I tell you a pop, I will tell you what it threw away.

1 What a Mixture Is

Parts that stay themselves when they are mixed

Trail mix, muddy water, a bowl of cereal and a pocketful of coins are all **mixtures**. A mixture has two or more parts that are put together but are not changed into something new. Each part is still what it was. Rice in a bowl of rice and lentils is still rice.

That means a mixture can usually be taken apart again. The big question of this book is how, and how to check that nothing went missing on the way.

Try it: Sort by hand

You need: a kitchen scale in grams, a bowl, three cups, about 100 g of rice, 60 g of lentils and 20 steel paper clips.

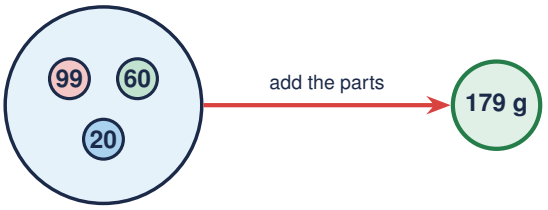
1. Weigh the rice, the lentils and the clips separately. Write each weight.
2. Put all three into the bowl and mix. Weigh the whole mixture. Write it.
3. Pick the three kinds apart by hand into the three cups.
4. Weigh each cup. Add the three weights. Compare the total with the mixture.

	Rice	Lentils	Clips	Whole mixture	Parts added
Before mixing (g)					
After sorting (g)					

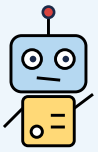
■ **Adult help:** Steel clips and small beans can be swallowed by little children and pets. Work at a table and put everything away afterwards. Do not taste anything.

Practice numbers, not from a real test

Whole mixture on the scale: 180 g.
After sorting: rice 99 g, lentils 60 g, clips 20 g. Total $99 + 60 + 20 = 179$ g.
The parts are 1 g short of the mixture. A grain of rice rolled off the table, and the scale itself can be 1 g off. We call 1 g *within the scale's error*.



kept: the weight of every part
set aside: how the mixture looked



After I sorted my mixture the parts weighed 1 gram less than the whole. So one gram of stuff stopped existing. I will write that down. Should I?

What makes this correction trustworthy?

What we checked. We weighed the whole mixture and each part, then added the parts and compared with the whole.

What this check gives us reason to trust. The parts came within 1 g of the whole, which is the size of the scale's own error.

What it does not show. That nothing at all was lost. A gap smaller than the scale's error cannot be told apart from no gap.

What did we keep?

We replaced a bowl of mixed things with three weights. **We kept** _____ **and we set aside** _____.

Puzzle 1.1: Add the parts

After sorting, rice was 98 g, lentils 61 g and clips 20 g. Total _____ g. The whole mixture weighed 180 g. How far off? _____ g

Puzzle 1.2: Mixture or not?

Write M for mixture and N for not a mixture.
(a) trail mix _____ (b) a spoon of pure salt _____ (c) muddy water _____ (d) one steel paper clip _____

Puzzle 1.3: Too big a gap

A 150 g mixture was sorted into 70, 50 and 25 g. Total _____ g. The gap is _____ g.
Is that within the scale's 1 g error? _____
What would you do first? _____

Puzzle 1.4: Spot the flaw

Sam weighed the whole mixture on a kitchen scale and the parts on a bathroom scale. The parts added to more. Why is this test not trustworthy?
_____

Puzzle 1.5: Still themselves

In a fruit salad, the grapes are still grapes. Is that true of every mixture? Explain.
_____

Puzzle 1.6: Draw the circles

The parts of a mixture were 40, 25 and 15 g. Fill in the empty circle with the total.



Puzzle 1.7: Biggest part

Rice 99 g, lentils 60 g, clips 20 g. Which part weighs the most? _____ Is it more than half of the 179 g? _____

Puzzle 1.8: Your own mixture

Name a mixture at home and its parts.

_____

Puzzle 1.9: Rounded weights

The scale showed 179 g for the parts. It showed 180 g for the whole. The smallest step on the scale is 1 g. Is the difference bigger than the smallest step? _____

Puzzle 1.10: Missing grain

A student says one grain of rice vanished. Write a more careful sentence.

_____

Mixture log

Look for mixtures at home. Name the parts and say how you could take them apart.

Mixture	Parts	How I could separate them

2 Sieves and Magnets

Using a difference between the parts

Sorting by hand works for a handful of rice. For sand or for a thousand grains it is too slow. Instead we look for something that is *different* about the parts and use a tool that notices that difference.

Tool	It notices a difference in	It works when
A sieve	size	one part is small enough to fall through
A magnet	steel or iron	one part is steel and the rest is not
Your hands	looks and shape	there are not many parts

Try it: Sieve the sand

You need: a kitchen scale, a sieve or tea strainer, a bowl, 150 g of sand and 60 g of lentils.

1. Mix the sand and the lentils. Weigh the whole mixture.
2. Hold the sieve over a bowl. Pour the mixture in and shake gently.
3. Weigh what fell through and what stayed on the sieve.

	Whole	Fell through	Stayed on sieve	Parts added
Weight (g)				

Practice numbers, not from a real test

Whole mixture: 210 g. Fell through: 148 g of sand. Stayed on: 60 g of lentils. $148 + 60 = 208$. The gap is $210 - 208 = 2$ g. Some sand stuck between the lentils, and a pinch dropped on the table.

Try it: Use a magnet

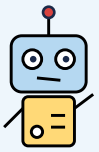
Mix 100 g of rice with 20 g of paper clips. Hold a magnet over the mixture, or pull it through. Weigh what it picks up and what is left.

	Whole	On the magnet	Left behind
Weight (g)			

■ **Adult help:** Small magnets can harm if swallowed. Keep them away from young children and pets. Some paper clips are plated and are not magnetic, so test one first.

Practice numbers

Whole: 120 g. The magnet picked up 18 g. The rest weighed 102 g. $18 + 102 = 120$. Two grams of clips stayed in the rice. They were a kind of clip that is not magnetic. The total still matched. Only the *split* was wrong.



A sieve has holes, so it can separate salt from sand. Table salt and sand grains are about the same size, but holes are holes. Will it work?

What makes this correction trustworthy?

What we checked. We weighed the whole and the parts after sieving and after the magnet, and we checked what stayed behind in each tool.

What this check gives us reason to trust. The totals matched within a couple of grams. The magnet check also showed us a hidden clip that the magnet could not pick up.

What it does not show. That a tool separates a mixture completely. A sieve or a magnet is only as good as the difference it relies on.

What did we keep?

We used a sieve and a magnet. **We kept** _____ **and we set aside** _____.

Puzzle 2.1: Choose the tool

Write S for sieve, M for magnet, H for hand or X if none of these will work.
(a) sand and pebbles _____ (b) rice and steel screws _____ (c) salt and sand _____
(d) sugar dissolved in tea _____ (e) flour and big stones _____

Puzzle 2.2: Add up

A sieve test started with 210 g. Sand through the sieve weighed 146 g and lentils on it 60 g. Total _____ g. Gap _____ g.

Puzzle 2.3: Magnet test

A magnet picked up 15 g from 90 g of rice and clips. The rest weighed 75 g. Add them. _____
Does it match? _____

Puzzle 2.4: Hidden clip

After the magnet, a student found a clip still in the rice. Which kind of clip is it? _____ Where did the gap come from? _____

Puzzle 2.5: Unfair test

Ana sieved damp sand and Ben sieved dry sand. Ana's sieve clogged. They say the sieve is bad. What is wrong with their test? _____

Puzzle 2.6: Order the steps

Number the steps for the sand and lentils from 1 to 4.
_____ Weigh what fell through. _____ Weigh the whole mixture.
_____ Add the two parts. _____ Shake the sieve.

Puzzle 2.7: Which property?

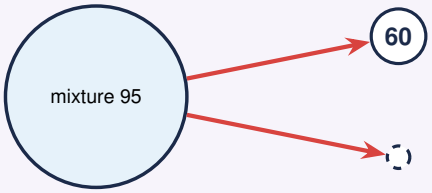
For each tool, write the property it uses.
Sieve: _____ Magnet: _____

Puzzle 2.8: Does the order matter?

Rice, clips and sand are mixed. Would you use the magnet first or the sieve first? Say why either order could work. _____

Puzzle 2.9: Draw the circles

A mixture of 60, 25 and 10 g is sieved into 60 and 35. Fill in the empty circle.



Puzzle 2.10: Design a sorter

Choose a mixture at home and say which difference between its parts you could use.
_____

Sieve record

Try three different mixtures through the same sieve. Weigh before and after, and record what stayed on and what fell through.

Mixture	Whole (g)	Fell through (g)	Stayed on (g)	Gap (g)

Which mixture split most cleanly? _____

Which left the biggest gap, and where might the gap be? _____

Sketch your sieve and mark where each part went.

3 Salt That Disappears

Dissolving, and what the scale says

Stir a spoonful of salt into water. The white grains vanish and the water is clear again. Where did the salt go? Is it gone for good?

Salt that has spread out among the water, too small to see, is **dissolved**. The water with the dissolved salt in it is a **solution**. A solution is a mixture whose parts you cannot see or sieve apart. But the parts are still there, and a scale can tell.

Try it: Weigh before and after

You need: a kitchen scale, a clear cup, water, table salt, a spoon.

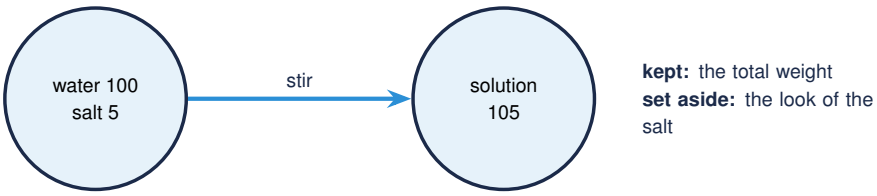
1. Put the cup on the scale and press the button that sets the scale to zero, or write the cup's weight. Pour in 100 g of water.
2. Put a small spoonful of salt on the scale in a saucer and weigh it. Write the weight.
3. Add the salt to the water without stirring. Weigh the cup, water and salt together.
4. Stir until the salt seems gone. Weigh again.

	Cup	Water	Salt	Total (g)
Before stirring				
After stirring				

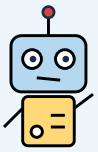
■ **Adult help:** Never taste anything from an experiment, not even the salt water. Wipe up spills.

Practice numbers, not from a real test

Cup 40 g. Water 100 g. Salt 5 g. Before stirring: $40 + 100 + 5 = 145$ g.
After stirring the cup still read 145 g. The salt looked gone, but its weight had not gone anywhere.



kept: the total weight
set aside: the look of the salt



The salt dissolved, so it vanished. The cup should weigh the same as 100 grams of water plus a cup. I weighed it myself and I trust my idea. Do the numbers agree?

What makes this correction trustworthy?

What we checked. We weighed the cup, the water and the salt before stirring and the whole cup after stirring.

What this check gives us reason to trust. The weight did not change when the salt dissolved. If the salt had stopped existing, the cup would have got lighter.

What it does not show. Where the salt is, or how much a spoonful of salt weighs for a different spoon. We only know that its weight stayed in the cup.

What did we keep?

We looked at salt in water. **We kept** _____ **and we set aside** _____.

Get the salt back

If the salt is still there, we should be able to get it back. Water leaves by evaporation (Book 5), so the salt should stay.

Try it: Evaporate the salt water

Weigh an empty saucer. Pour about 25 g of salt water into it. Put it on a sunny sill for a few days. When it looks dry, weigh the saucer again.

Empty saucer (g)	
Saucer after the water has gone (g)	
Crust of salt (g)	

Practice numbers

Empty saucer: 30 g. After three sunny days: 33 g. Crust: $33 - 30 = 3$ g.
The crust was white and hard, and it was the salt left behind when the water evaporated.

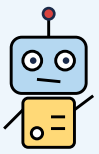
How much will dissolve?

Try it: Spoonfuls till it stops

Add level spoonfuls of salt to 100 g of water, stirring until each is gone. Count the spoonfuls until some stays at the bottom. How many did you add? _____

Practice numbers

5 spoonfuls dissolved. The 6th stayed as grains at the bottom. The water had taken in as much salt as it could.



Clear water must be pure water. Salt water looks clear, so I can tell it is plain water. Is that true?

What makes this correction trustworthy?

What we checked. We got the salt back by evaporation and weighed the crust.

What this check gives us reason to trust. A solid crust of salt remained, so something solid really was dissolved in the clear water.

What it does not show. That we got back every gram. Some crust sticks to the saucer, and the scale is only exact to about 1 g.

Puzzle 3.1: Total weight

A cup weighs 40 g, the water 100 g and the salt 7 g. What does the cup weigh after the salt dissolves? _____ g

Puzzle 3.2: Crust

An empty saucer weighs 28 g. After the salt water dried it weighs 33 g. How much salt was left? _____ g

Puzzle 3.3: Clear is not pure

Two cups of clear water look the same. One has dissolved salt. Without tasting, how could you find out which? _____

Puzzle 3.4: Spoonfuls

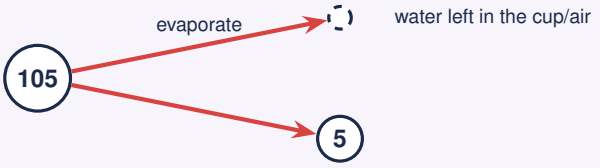
In a practice test 5 spoonfuls dissolved and the 6th did not. Write the sentence that fits the data: _____

Puzzle 3.5: Spot the unfair test

Sam added salt to cold water and Mia added salt to warm water. Mia's salt dissolved quicker. Which two things differ? _____

Puzzle 3.6: Draw the circles

The solution weighs 105 g. The salt was 5 g. Fill in the circle.



Puzzle 3.7: Sentence check

Mark S if the data supports it, G if it is a guess about why, and B if it is too big for the data.
(a) "The cup weighed the same after stirring." _____
(b) "The salt broke into tiny pieces." _____
(c) "Nothing ever disappears." _____

Puzzle 3.8: Another solid

Name another solid that might dissolve in water, and one that might not. Say how you could find out without tasting. _____

Puzzle 3.9: Short on the scale

After the salt water dried, the crust weighed 2 g, but the salt you added weighed 3 g. Name two reasons for the missing gram. _____

Puzzle 3.10: Your test

Plan a fair test of whether stirring faster dissolves salt sooner. Change: _____
Keep the same: _____ Measure: _____

Salt crust diary

Keep a record of getting the salt back. Weigh the saucer each day.

Day	Saucer and contents (g)	What it looks like	Crust so far (g)
0			
1			
2			
3			
4			
5			

When did the weight stop dropping? _____

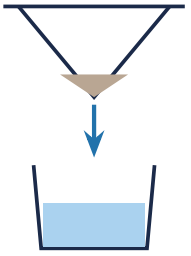
Why does it stop? _____

Draw the saucer on the first and last day.

4 Muddy Water Through Paper

Catching what is too big to pass

Muddy water is a mixture of water and tiny bits of soil. A sieve is no good, because the bits are too small. A **filter** is paper with very tiny holes. Water passes through the holes. The soil bits are too big and are caught.



muddy water goes in

mud is caught on the paper

clearer water comes out

Try it: Filter muddy water

You need: a kitchen scale, a coffee filter or paper towel, two cups, a spoon, a little garden soil, water.

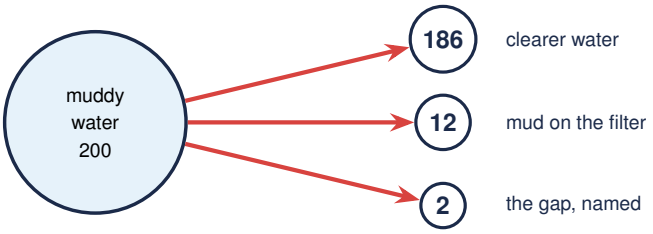
1. Weigh a dry filter. Write it.
2. Stir about 10 g of soil into 190 g of water in a cup, to make 200 g of muddy water. Write 200.
3. Hold the filter over a clean cup. Pour the muddy water slowly into the filter.
4. Weigh the clear water collected. Then weigh the wet filter with the mud on it.

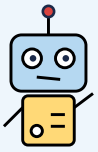
Muddy water at the start (g)	
Dry filter (g)	
Clear water collected (g)	
Wet filter with mud (g)	
Mud and water on the filter (wet minus dry) (g)	

■ **Adult help:** Use garden soil that is free of sharp bits. Wash hands afterwards. Do not drink the water.

Practice numbers, not from a real test

Muddy water: 200 g. Dry filter: 2 g. Clear water collected: 186 g. Wet filter with mud: 14 g.
On the filter: $14 - 2 = 12$ g of mud and water.
Add up: $186 + 12 = 198$ g. The gap is $200 - 198 = 2$ g. Some water stayed in the cup and on the sides.





The water coming through looks clean, so the filter must have made it pure. It looks clear to me. I like being sure. Should I be?

What makes this correction trustworthy?

What we checked. We weighed the muddy water, the clear water and the wet filter, and named the small gap.

What this check gives us reason to trust. The three amounts added to within 2 g of the start, so the mud really stayed on the filter rather than disappearing.

What it does not show. That the clear water is pure. A filter catches what is too big for its holes. Dissolved salt is tiny and would pass straight through.

What did we keep?

We filtered muddy water. **We kept** _____ **and we set aside** _____.

Does the filter catch salt?

Try it: Muddy salt water

Dissolve 5 g of salt in 100 g of water, add 5 g of soil, stir and filter. Put 25 g of the filtered water on a weighed saucer and let it dry. Is there a crust? Weigh it.

Empty saucer _____ g Dry saucer _____ g Crust _____ g

Practice numbers

Empty saucer: 30 g. Dry saucer: 33 g. Crust: 3 g. The filtered water was clear, yet a crust of salt remained. The salt had passed through the filter.

Which tool for which difference?

Tool	Catches	Lets through
Sieve	big bits	small bits and water
Filter paper	soil, sand, bits you can see or cloud	water with dissolved salt
Evaporation	nothing (water leaves)	salt stays behind

Puzzle 4.1: Add up the filter

Muddy water 200 g. Clear water 185 g. Mud and water on the filter 13 g. Total _____ g. Gap _____ g.

Puzzle 4.2: Wet minus dry

A dry filter weighs 2 g. After filtering it weighs 17 g. How much mud and water is on it? _____ g

Puzzle 4.3: Choose the tool

Write S for sieve, F for filter, E for evaporation or X if none will work.
(a) sand from water _____ (b) salt from clear salt water _____ (c) pebbles from sand _____

Puzzle 4.4: Clear but not pure

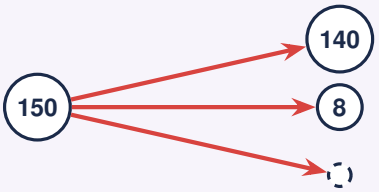
A filter turned muddy water clear. A student says it is now pure. How could she test?
_____

Puzzle 4.5: Spot the unfair test

Tom filtered muddy water through a coffee filter. Lena filtered it through a paper towel. Lena's water was cloudier, so she says muddy water cannot be filtered. What is wrong?
_____

Puzzle 4.6: Draw the circles

Muddy water of 150 g gave 140 g of clearer water and 8 g on the filter. Fill in the gap.



Puzzle 4.7: Is the gap too big?

The gap in the last puzzle is _____ g. Is that within the scale's 1 g error? _____
What would you check? _____

Puzzle 4.8: Sentence check

Mark S, G or B.
(a) "The filtered water weighed 186 g." _____
(b) "The filter caught all the soil." _____
(c) "Filtering makes any water safe." _____

Puzzle 4.9: Cloudy water

A cup of cloudy water was left to stand overnight. The next day the bottom of the cup was muddy. Explain. _____

Puzzle 4.10: Your test

Plan a fair test of which paper filters best: coffee filter, paper towel or tissue. Change: _____ Keep the same: _____ Measure: _____

Filter comparison

Filter the same amount of the same muddy water through two or three different papers. Keep everything else the same.

Paper	Muddy water in (g)	Clear water out (g)	Left on paper (g)	How clear?

Which paper was clearest? _____

Which was quickest? _____

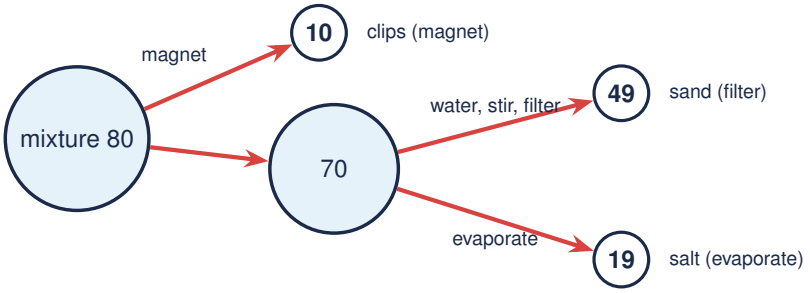
Is the quickest also the clearest? Say what that tells you. _____

Draw the cups side by side to show how clear each was.

5 Separating Everything

Putting the tools in the right order

Now we can use all the tools together. Here is a mixture of three parts: steel paper clips, sand and salt. No single tool can take it apart, but the tools one after another can.



Try it: Sand, salt and clips

You need: a kitchen scale, a magnet, a filter, a saucer, three cups, water, 10 g of paper clips, 50 g of sand and 20 g of salt.

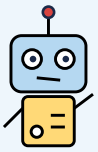
1. Mix the clips, sand and salt. Weigh the mixture.
2. Pull the clips out with the magnet. Weigh the clips and what is left.
3. Stir the rest into 100 g of water so that the salt dissolves. The sand will not dissolve.
4. Filter it. The sand stays on the filter. Dry the sand on a plate and weigh it.
5. Pour the salty water that came through onto a weighed saucer and let it evaporate. Weigh the salt crust.

	Whole	Clips	Sand	Salt	Parts added
Weight (g)					

■ **Adult help:** Do not taste anything. Keep small magnets and paper clips away from young children and pets. Wash hands after handling sand and salt water.

Practice numbers, not from a real test

Whole: $10 + 50 + 20 = 80$ g. After the magnet: clips 10 g, left 70 g.
Sand, dried: 49 g. Salt crust: 19 g.
Parts added: $10 + 49 + 19 = 78$ g. The gap is 2 g. We weighed three separate parts, so the scale's 1 g error could add up to 3 g, and a 2 g gap fits.



I will filter the dry mixture first, because the filter will keep the sand and the salt together. Then I will separate them. That is quicker, no? Did I miss something?

What makes this correction trustworthy?

What we checked. We weighed the whole mixture and each part, and we did the steps in an order where each tool had a difference to use.

What this check gives us reason to trust. The parts added to 78 g against 80 g, a gap smaller than the combined scale error.

What it does not show. That a bigger gap would be harmless, or that we can tell which step lost what from only the final weights. The next chapter shows how to find the step.

What did we keep?

We separated three parts. **We kept** _____ **and we set aside** _____.

Puzzle 5.1: Order the steps

Number these from 1 to 6.
_____ Add water and stir. _____ Weigh the whole mixture. _____ Evaporate the salt water.
_____ Pull out the clips with a magnet. _____ Filter the mixture. _____ Weigh the dry sand.

Puzzle 5.2: After the magnet

The mixture weighs 80 g. The magnet picks up 10 g of clips. How much is left? _____ g

Puzzle 5.3: Add the parts

Clips 10 g, sand 49 g, salt 19 g. Total _____ g. Whole mixture 80 g. Gap _____ g.

Puzzle 5.4: Why dissolve first?

Why must the salt dissolve before filtering? _____

Puzzle 5.5: Which tool?

For each part, write the tool: magnet, filter or evaporation.
Clips: _____ Sand: _____ Salt: _____

Puzzle 5.6: Where did the sand lose a gram?

You started with 50 g of sand and ended with 49 g. Give two places the missing gram could be. _____

Puzzle 5.7: Where did the salt lose a gram?

Starting with 20 g of salt, the crust weighed 19 g. Give two places the missing gram could be. _____

Puzzle 5.8: A bigger gap

Another group's parts added to 71 g from a mixture of 80 g. Is a 9 g gap within error? _____ What would you do? _____

Puzzle 5.9: Rice, salt and clips

Plan how to separate rice, salt and clips. List the steps in order.

_____

_____

Puzzle 5.10: Spot the unfair test

Group A dried their sand for one hour and Group B for a whole day. Group B's sand weighed 3 g less. Why? _____

Plan your own separation

Pick a mixture of three parts, such as rice, salt and paper clips, or lentils, sand and salt. Plan the steps before you start, then record what happened.

My mixture and its parts: _____

The difference each tool will use:

_____

_____

Step	What I will do	Expected (g)	Weighed (g)
1			
2			
3			
4			
5			

Total gap: _____ g Where do I think it went? _____

6 Finding What Went Missing

Weighing after every step

A ledger at the end tells you *that* something went missing. It cannot tell you *where*. To find the step that lost something, weigh after **every** step and compare each result with what you expected.

Model	What it keeps	What it drops
A mixture as parts	the weight of each part	how it looked
A sieve or magnet	the split by size or by steel	bits that get stuck
A solution	the total weight	where the salt sits
A filter	what is too big to pass	anything dissolved
A step ledger	where the loss happened	the cause of the loss

Try it: A step ledger

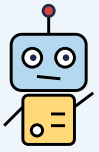
Make a mixture of 12 g clips, 30 g pebbles and 58 g sand (100 g in all). Separate it in two steps: magnet, then sieve. Weigh the mixture and every product after each step. Fill in what you expected and what you found.

After	Expected (g)	Weighed (g)	Gap (g)
Start	100		
Magnet: clips out, rest left			
Sieve: pebbles on it, sand through			

Practice numbers, not from a real test

Start: 100 g. After the magnet: clips 12 g and rest 88 g. Expected 88, weighed 88. No gap. After the sieve: pebbles 30 g on the sieve, sand through into the bowl 53 g. Expected 58, weighed 53. The gap is 5 g.
The gap appeared at the *sieve step*. A little pile of sand was on the table beside the bowl.

After	Expected	Weighed	Gap
Start	100	100	0
Magnet	88	88	0
Sieve (sand through)	58	53	**5**



The total was 5 grams short at the end, so 5 grams of sand turned to dust and vanished. A loud thought, but a good one. Is it?

What makes this correction trustworthy?

What we checked. We weighed after every step, so each step could be checked against what it should have produced.

What this check gives us reason to trust. Only the sieve step showed a gap, and the pile of sand on the table matched its size.

What it does not show. That every small gap has such a visible cause. Gaps of 1 g or less are within the scale's error and cannot be traced.

What did we keep?

We turned a separation into a table of weights. **We kept** _____ **and we set aside** _____.

Puzzle 6.1: Find the step

Start 80 g. After step 1 expected 70, weighed 70. After step 2 expected 49, weighed 44. At which step did the gap appear? _____ How big? _____ g

Puzzle 6.2: Fill the table

Start 90 g. The magnet takes out 15 g. What do you expect to be left? _____ g

Puzzle 6.3: Within error?

A gap of 1 g appeared at a step weighed on a scale with 1 g error. Can you tell where it went? _____ Why or why not? _____

Puzzle 6.4: Check one step

Why is weighing after every step better than weighing only at the end? _____

Puzzle 6.5: Clue

A 5 g gap appeared at the sieve. Name two clues you would look for. _____

Puzzle 6.6: Add the table

A table shows gaps of 0, 2 and 1 g after three steps. Total gap? _____ g

Puzzle 6.7: Model or real?

Write M for model and R for real thing.
(a) the table of weights _____ (b) the pile of sand on the table _____ (c) a circle with 88 in it _____

Puzzle 6.8: Which question?

The ledger can answer "where did the weight go?" Can it answer "why did the sand spill?" _____ What else would you need? _____

Puzzle 6.9: Unfair ledger

Ben weighed after the magnet on one scale and after the sieve on another. Why can that mislead? _____

Puzzle 6.10: Recount

Clips 12 g, pebbles 30 g and sand 53 g. Total _____ g

Puzzle 6.11: Wrap

The student wrote “5 g of sand vanished”. Wrap it with a better claim. _____

Puzzle 6.12: A good habit

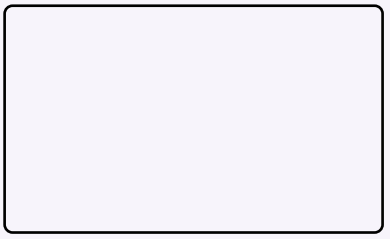
Write one habit from this book that makes a mixture test more trustworthy. _____

Puzzle 6.13: Missing step

A student weighed the whole, then the clips, then the sand, and then said the salt was “what is left”. What could go wrong? _____

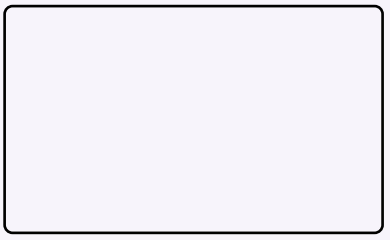
Puzzle 6.14: Draw the step circles

Draw the magnet step as circles: a whole of 90 g, clips of 15 g, and the rest.



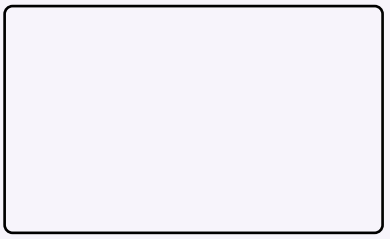
Puzzle 6.15: Draw the gap

Draw the sieve step with a 5 g gap as its own circle.



Puzzle 6.16: Draw the wrap

Draw a circle around “5 g vanished” and write the corrected claim outside it.

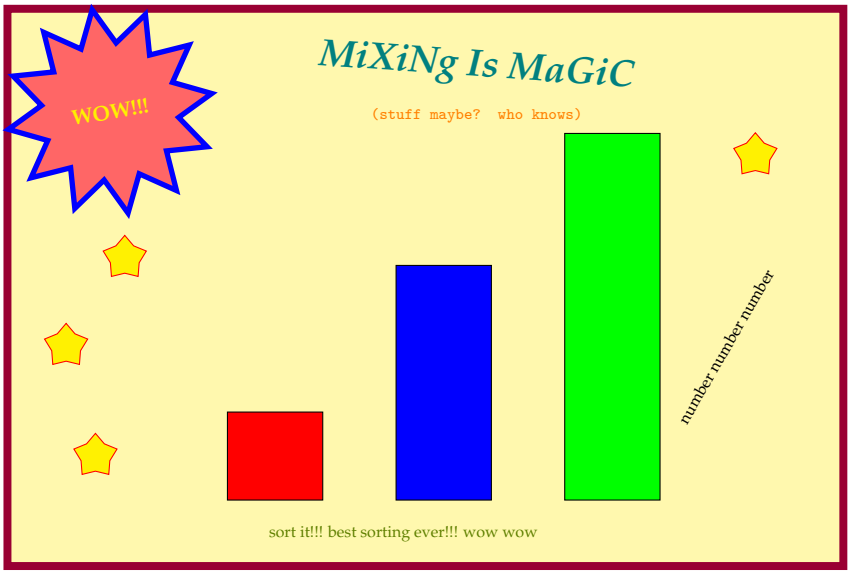


Slop check

Slop is hasty, gaudy work that forgot to take care. It can be made by a person or a machine, quickly or slowly. What matters is whether the work shows care for three things:

- **Salient:** Is the main point easy to see?
- **Simple:** Is there anything that does nothing but get in the way?
- **Relevant:** Does everything on the page help answer the question?

Here is a hasty poster about a separation. Study it, then decide what is wrong.



Slop check

(a) What is the poster's main point? Can you tell?

_____

(b) Name two things that are not simple, because they only distract.

_____

(c) Name two things that are missing, so the bars cannot be read.

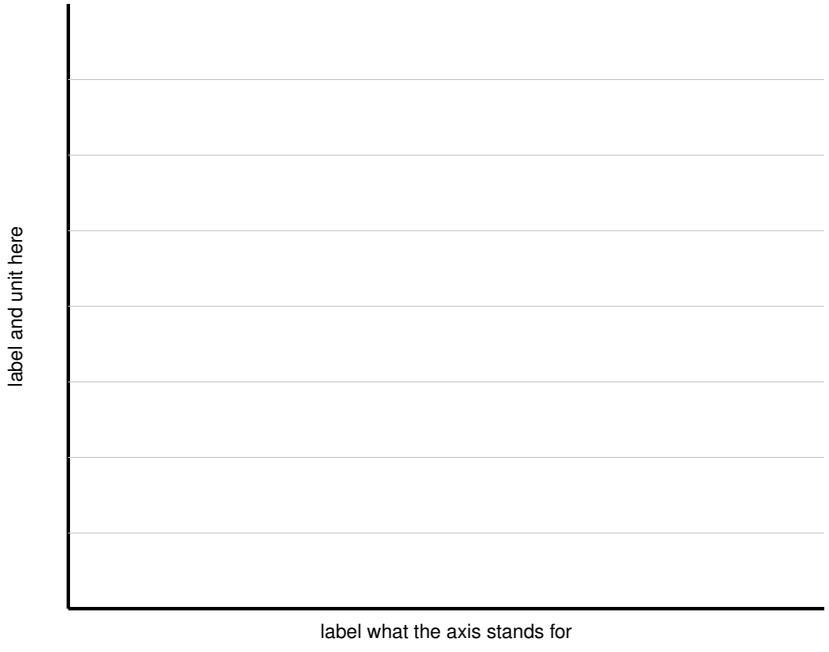
_____

(d) Which part is not relevant to mixtures at all?

_____

Now you try

On a separate sheet, redraw a clear bar chart of the three parts of the mixture: clips 10 g, sand 49 g and salt 19 g. Give it a title, label the axes with units, use one or two calm colours, and delete everything that does not help. Check it against the three words: salient, simple, relevant.

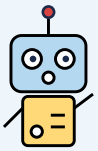


Title:

What did we keep, across the whole book?

For each chapter, say one thing that was simplified, what was kept, and what was set aside. A sample is done for you.

Chapter	What we simplified	What we kept	What we set aside
1 What a mixture is	a bowl of things as weights	the weight of each part	how it looked
2 Sieves and magnets	a tool as one difference	the split by size or steel	bits that got stuck
3 Salt that disappears	dissolved salt as a weight	the total weight	where the salt sits
4 Muddy water	a filter as a catcher	the mud's weight	dissolved things
5 Separating everything	a plan as ordered steps	the order and the total	small losses
6 Missing weight			



Everything you weighed in this book was a simplification of something. A model is allowed to drop things, as long as you say which. Next time something seems to vanish, you will know to weigh again.

Write a separation report

1. **What I looked at.**
2. **What I changed, measured and kept the same.**
3. **My readings.** All of them.
4. **My answer.** With the gap named.
5. **What I kept and what it hides.**

Pick one separation from this book and write a report on it. Use the lines below.

Tests to try

Pick one and design it with the habits from this book: change one thing, keep the rest the same, repeat, and weigh every step.

1. Does warm water dissolve more salt than cold water?
2. Does stirring change how fast sugar dissolves?
3. Which paper filters muddy water clearest: coffee filter, paper towel or cloth?
4. Can you separate rice, lentils and sand with a sieve alone?
5. How much of a salt crust can you get back from 100 g of salt water?
6. Which kitchen mixture is easiest to take apart, and which is hardest?

Write the test you pick:

_____

_____

_____

_____

_____

_____

Change: _____ Measure: _____ Keep the same: _____

A week of mixtures

Each day, notice one mixture and say how you could take it apart.

Day	The mixture	A tool I could use
Mon		
Tue		
Wed		
Thu		
Fri		
Sat		
Sun		

Questions I still have

Write what you still wonder about mixtures. Then pick one and think how to test it.

Answers

Try every puzzle first. Where an answer shows a calculation, check the steps, not just the result.

Chapter 1

Multix on the missing gram. No. A gap of 1 g is the size of the scale's error, so it does not show that anything stopped existing. **Keep box.** Kept: the weight of each part. Set aside: how the mixture looked.

1.1 $98 + 61 + 20 = 179$ g. Off by 1 g.
1.2 (a) M (b) N (c) M (d) N.
1.3 $70 + 50 + 25 = 145$, a gap of 5 g. No. Reweigh, check for spills and repeat on one scale.
1.4 Two scales can disagree by a few grams. Use one scale for every weighing.
1.5 In a mixture the parts keep their own character, even when dissolved parts cannot be seen. Chapter 3 shows this.
1.6 $40 + 25 + 15 = 80$.
1.7 Rice. 99 is more than half of 179, since half is 89.5.
1.8 Answers vary.
1.9 No. The difference is 1 g, the same as the smallest step.
1.10 "One grain probably rolled off the table, and the scale may be 1 g off."

Chapter 2

Multix on salt and sand. No. Salt grains and sand grains are about the same size, so the holes do not tell them apart. **Keep box.** Kept: the split by size or by steel. Set aside: bits that stuck.

2.1 (a) S (b) M (c) X (d) X (e) S.
2.2 $146 + 60 = 206$, a gap of 4 g.
2.3 $15 + 75 = 90$. Yes.
2.4 A clip that is not magnetic, such as a plated or stainless one. The total matched, but the split was wrong.
2.5 One group used damp sand and one used dry, so the sand differed. Damp sand clogs a sieve.
2.6 Weigh the whole 1, shake the sieve 2, weigh what fell through 3, add the parts 4.
2.7 Sieve: size. Magnet: steel or iron.
2.8 Either order works, since each tool ignores the other parts. Magnet first keeps clips from catching in the sieve.
2.9 35.
2.10 Answers vary.

Chapter 3

Multix on the weights. The cup weighs the same as before stirring, so it weighs more than 100 g of water plus the cup. The salt's weight stayed. **Keep box.** Kept: the total weight. Set aside: where the salt sits.

3.1 $40 + 100 + 7 = 147$ g.
3.2 $33 - 28 = 5$ g.
3.3 Put a spoonful of each on a saucer and let it dry. The salty one leaves a crust.
3.4 "Five spoonfuls dissolved in 100 g of water and the sixth stayed as grains."
3.5 The water temperature and the person who added the salt (and stirred).
3.6 The empty circle is 100, the water that leaves as vapour.
3.7 (a) S (b) G (c) B.
3.8 Sugar dissolves and sand does not. Stir and look for grains, or let it dry and look for a crust.
3.9 Some crust sticks to the saucer, and the scale is only exact to about 1 g.
3.10 Answers vary.

Chapter 4

Multix on pure water. No. A filter catches what is too big to pass. Dissolved salt would pass through. **Keep box.** Kept: the mud's weight. Set aside: anything dissolved.

4.1 $185 + 13 = 198$, a gap of 2 g.
4.2 $17 - 2 = 15$ g.

4.3 (a) F (b) E (c) S.
4.4 Dry a spoonful on a saucer. A crust means something was dissolved.
4.5 The filters were different papers. The paper towel may have bigger holes. "Cannot be filtered" is too big a claim.
4.6 $150 - 140 - 8 = 2$.
4.7 2 g. No. Check the cup and funnel for water and reweigh.
4.8 (a) S (b) G (c) B.
4.9 The mud is heavier than the water and settles to the bottom over time.
4.10 Answers vary.

Chapter 5

Multix on filtering first. The salt is dissolved only after water is added, so a dry mixture cannot be filtered apart. Dissolve first. **Keep box.** Kept: the order of the steps and the total. Set aside: small losses.

5.1 Weigh the whole 1, magnet 2, add water and stir 3, filter 4, then evaporating the salt water 5 and weighing the dry sand 6 (5 and 6 may swap).
5.2 $80 - 10 = 70$ g.
5.3 $10 + 49 + 19 = 78$, a gap of 2 g.
5.4 Dissolved salt passes through the filter. Undissolved salt would stay with the sand.
5.5 Clips: magnet. Sand: filter. Salt: evaporation.
5.6 Sand stuck to the filter or the cup, or a pinch dropped.
5.7 Salt crust stuck to the saucer, or salt water left in the cup.
5.8 No. 9 g is much more than the scale's error. Retrace the steps and weigh after each.
5.9 Magnet for the clips, add water and dissolve the salt, sieve or pick out the rice, evaporate the salt water.
5.10 The sand was drier after a day, so less water was left in it.

Chapter 6

Multix on the dust. No. The ledger showed the gap at the sieve, and a pile of sand on the table matched. **Keep box.** Kept: where the loss happened. Set aside: the cause of the loss.

6.1 Step 2. 5 g.
6.2 $90 - 15 = 75$ g.
6.3 No. A 1 g gap is within the scale's error, so it cannot be traced.
6.4 Weighing after every step shows the step where a gap appeared.
6.5 A pile of sand on the table, sand stuck in the sieve, or sand in the bowl.
6.6 $0 + 2 + 1 = 3$ g.
6.7 (a) M (b) R (c) M.
6.8 No. The ledger shows where, not why. You would need to watch or repeat the pour.
6.9 Different scales can disagree, so the gap could come from the scales.
6.10 $12 + 30 + 53 = 95$ g.
6.11 "5 g of sand went missing at the sieve step and was found on the table."
6.12 Weigh after every step, use one scale, or name each gap.
6.13 Calling the leftover "salt" hides any loss. You should weigh the salt.
6.14 to 6.16 Answers vary. Check that every circle's weight is accounted for.
Slop check. (a) You cannot tell: no title, no labels. (b) Many colours, stars, joke text. (c) A title, the parts, and the weights with units. (d) The stars and jokes.

Words

dissolve to spread out in a liquid until the parts cannot be seen.

evaporate to turn from liquid into vapour and leave.

filter paper with tiny holes that lets liquid pass but catches bits.

ledger a list of weights that adds up to a total.

mixture two or more things put together that keep their own character.

model a simple stand-in that keeps some things and drops others.

sieve a net or screen with holes that lets small bits fall through.

solution a mixture of a liquid with something dissolved in it.

Notes for grown-ups

This book teaches that a separation is checked by a ledger of weights: the parts, added up, should match the whole within the scale's error, and weighing after every step shows where a gap appeared. Each chapter has one activity, one worked example with made-up practice numbers, one trust box, one "what did we keep?" prompt and a page of puzzles.

Materials and safety. A kitchen scale in grams, magnets, steel paper clips, sand, salt, rice, lentils, soil, filters, cups and a saucer. Never taste an experiment. Keep small magnets and paper clips away from young children and pets. Some clips are plated and not magnetic.

Scale error. A kitchen scale reading to 1 g can be off by about 1 g. The book treats gaps of that size as unexplained and larger gaps as worth tracing.

The practice numbers were made up so the arithmetic is easy to check. Real results will vary with the equipment and the materials.

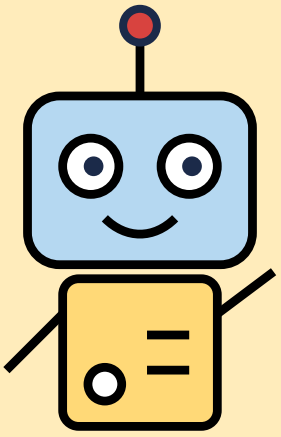
Sources

- T. L. Brown, H. E. LeMay and others. *Chemistry: The Central Science*. Pearson. For mixtures, solutions and methods of separation.
- J. R. Taylor. *An Introduction to Error Analysis*, 2nd edition. University Science Books, 1997. For measurement error and repeats.

Next in Science

Sorting the Living

A tray holds leaves, shells and buttons. Which belong together? Book 7 builds a yes-or-no key, trades it with a friend, and tests whether it sorts the same way twice.



Ages 9 to 11 ⋆ **Science Book 6**